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CALCULATION OF THE CHARACTERISTICS OF TRAVELING WAVES IN LAYERED MEDIA

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A new method is considered for the calculation of traveling-wave characteristics in a layered medium. It is mainly based on the transfer-matrix method [2–10] and the generalized reflection-transmission coefficient (GRTC) method [11–14] for the calculation of characteristics. Introducing an impedance tensor of rank 2, $\hat{Z}(z)$, in the boundary-value stress problem and utilizing in this way the boundary conditions and the Lame equation, we obtain a system of differential equations which is solved by the fourth-order Runge-Kutta method. Specifying a model of the layered medium in terms of known constants, we find the traveling-wave characteristic γ as a function of frequency ω. Given γ, we easily calculate the dispersion curves, which closely fit the GRTC method. Compared with the GRTC method, the impedance-tensor method fully solves the dispersion curve of the normal-mode traveling waves, and it is applicable to both a homogeneous planar Earth model and a model with a slow layer.

Keywords: traveling wave characteristics, impedance-tensor method, dispersion curve, phase velocity, group velocity

Introduction

Focused studies of the structure of the Earth's interior have produced some important results in recent years [1–8]. In particular, the velocity structure of transverse waves is now mainly investigated using surface-wave dispersion formulas [9–12]. Solving the elastic-wave equation is one of the most significant scientific directions in seismology (for instance, investigation of the characteristics of seismic phases, calculation of seismic wave paths, calculation of synthetic seismograms, and so on.). Different waves have different characteristics. The determination of these characteristics plays a major role in the solution of the wave equation and the calculation of wave dispersion.

Most studies of the forward seismic-wave problem compute the plane-wave dispersion. The classical method for a piecewise-constant plane-layered medium is the Thomson–Haskell transfer-matrix method [2]. However, the numerical computations involve loss of high-frequency accuracy. A number of new methods have been developed to overcome this difficulty [3–10]. X. F. Chen [11] has proposed the generalized reflection-transmission coefficient (GRTC) method [11]. The impedance tensor method is often used in magnetotelluric sounding. Dmitriev [1] applies this method in seismology to solve for the spectral characteristics in an isotropic layered medium. In the present article, we consider a new impedance-tensor method to calculate the traveling-wave characteristics from seismic data.

Calculation of Traveling-Wave Characteristics

Consider a layered medium with the boundary $z = z_n$, $n \in [0, n]$, where $z_0 = 0$ is the Earth's surface. Inside each layer, $z \in [z_{n-1}, z_n]$, $n \in [1, N]$. The seismic parameters λ_n, μ_n, ρ_n are assumed constant.

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In this layered medium, a traveling wave propagates in the direction of the OX axis. The displacement vector in the traveling wave is specified in the form

$$\vec{U}(x,y,z) = \overline{u}(z)e^{i\gamma x+i\omega t}, \tag{1}$$

where ω is the seismic-field frequency and γ is the traveling-wave propagation constant (characteristic); $\overline{u}(z)$ is the traveling-wave amplitude. Inside the layer, $\overline{u}(z)$ satisfies the Lame equation with constant coefficients:

$$(\lambda_n + 2\mu_n)\text{grad}\,\text{div}\vec{U} - \mu_n \text{rot}\,\text{rot}\vec{U} + \omega^2\rho_n\vec{U} = 0, \quad z \in [z_{n-1}, z_n]. \tag{2}$$

At the layer interface, the displacement $\overline{U}(z)$ and the stress $\{\sigma_{ij}\}$, $i, j \in [x,y,z]$, are continuous. Here

$$\sigma_{ij} = \lambda\delta_{ij}\,\text{div}\,\overline{U}(z) + \mu\left(\frac{\partial\overline{U}(z)}{\partial x_j} + \frac{\partial\overline{U}(z)}{\partial x_i}\right), \quad \{x_i\} = \{x,y,z\}, \quad i \in [1,3]. \tag{3}$$

Substituting the traveling-wave expression (1) in the Lame equation and the expression for the stress, we obtain

$$\begin{cases} (\lambda+2\mu)i\gamma\left(i\gamma u_x(z) + u_z'(z)\right) - \mu\left(-u_x'' + i\gamma u_z'\right) + \omega^2\rho u_x = 0, \\ -\mu\left(u_y'' - \gamma^2 u_y\right) + \omega^2\rho u_y = 0, \\ (\lambda+2\mu)\left(i\gamma u_x'(z) + u_z''(z)\right) - \mu\left(i\gamma\left(u_x' - i\gamma u_z\right)\right) + \omega^2\rho u_z = 0, \end{cases} \tag{4}$$

$$\sigma_{xz} = \mu\left(i\gamma u_z(z) + u_x'(z)\right), \tag{5}$$

$$\sigma_{zz} = \left(\lambda + 2\mu\right)u_z'(z) + \lambda i\gamma u_x(z), \tag{6}$$

The polarization of the traveling wave is independent of the coordinate Oy, and thus $u_y(z) = 0$. As a result, we obtain a system of two equations for the displacements u_x, u_z. Simplifying (1.4), we have

$$\begin{cases} u_x'' + a_x^2 u_x + b_1 u_z' = 0, \\ u_z'' + a_z^2 u_z + b_2 u_x' = 0, \end{cases} \tag{7}$$

where

$$a_x^2 = \frac{\rho\omega^2 - (\lambda+2\mu)\gamma^2}{\mu}, \quad b_1 = \frac{(\lambda+\mu)i\gamma}{\mu}, \quad a_z^2 = \frac{\rho\omega^2 - \mu\gamma^2}{\lambda+2\mu}, \quad b_2 = \frac{(\lambda+\mu)i\gamma}{\lambda+2\mu}. \tag{8}$$

As $z \to \infty$, u_x, $u_z \to 0$, we have to ensure that $\operatorname{Im} a_x \neq 0$ and or $\operatorname{Im} a_z \neq 0$,

$$a_x = i\sqrt{\frac{(\lambda+2\mu)\gamma^2 - \rho\omega^2}{\mu}}, \quad a_z = i\sqrt{\frac{\mu\gamma^2 - \rho\omega^2}{\lambda+2\mu}}. \tag{9}$$

System (7) was solved [1] by specifying the system of differential equations in the vector form:

$$\frac{d\overline{V}}{dz} = \hat{A}\overline{V}, \tag{10}$$

where

$$\overline{V} = \left(v_1 = u_x, v_2 = u_z, v_3 = u_x', v_4 = u_z'\right),$$

$$\hat{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a_x^2 & 0 & 0 & -b_1 \\ 0 & -a_z^2 & -b_2 & 0 \end{pmatrix}. \tag{11}$$

Assume that the differential equation (10) has the general solution $\overline{V} = \overline{Y}e^{\eta z}$. The necessary and sufficient condition has the form

$$\det(\hat{A} - \eta E) = \det\begin{pmatrix} -\eta & 0 & 1 & 0 \\ 0 & -\eta & 0 & 1 \\ -a_x^2 & 0 & -\eta & -b_1 \\ 0 & -a_z^2 & -b_2 & -\eta \end{pmatrix} = 0. \tag{12}$$

Find the roots:

$$\begin{aligned} \eta_1 &= \sqrt{\gamma^2 - \frac{\omega^2\rho}{\mu}}, \quad \eta_3 = -\sqrt{\gamma^2 - \frac{\omega^2\rho}{\mu}}, \\ \eta_2 &= \sqrt{\gamma^2 - \frac{\omega^2\rho}{(\lambda+2\mu)}}, \quad \eta_4 = -\sqrt{\gamma^2 - \frac{\omega^2\rho}{(\lambda+2\mu)}}. \end{aligned} \tag{13}$$

The four roots η are the eigenvalues of the matrix $\hat{A}$. Therefore, for the matrix $\hat{A}$ we obtain four eigenvectors in the form

$$a_1 = \left[-i\eta_1, -\gamma, -i\eta_1^2, -\gamma\eta_1\right]^T,$$

$$a_2 = \left[-i\eta_1, \gamma, i\eta_1^2, -\gamma\eta_1 \right]^T,$$

$$a_3 = \left[\gamma, -i\eta_2, \gamma\eta_2, -i\eta_2^2 \right]^T,$$

$$a_3 = \left[\gamma, i\eta_2, -\gamma\eta_2, -i\eta_2^2 \right]^T.$$

Then (10) has the following solution:

$$\bar{V} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = (a_1, a_2, a_3, a_4)\,\mathrm{diag}\,(e^{\eta_1 z}, e^{-\eta_1 z}, e^{\eta_2 z}, e^{-\eta_2 z}) \begin{pmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{pmatrix}. \tag{14}$$

The x- and z-components of the displacement field $\vec{U}$ can thus be represented as

$$\begin{cases} u_x = -i\eta_1 e^{\eta_1 z} C_1 - i\eta_1 e^{-\eta_1 z} C_2 + \gamma e^{\eta_2 z} C_3 + \gamma e^{-\eta_2 z} C_4, \\ u_z = -\gamma e^{\eta_1 z} C_1 + \gamma e^{-\eta_1 z} C_2 - i\eta_2 e^{\eta_2 z} C_3 + i\eta_2 e^{-\eta_2 z} C_4. \end{cases} \tag{15}$$

Substituting (13) in the continuity condition for the stress (3), we obtain

$$\begin{cases} \sigma_{xz} = -P_1 e^{\eta_1 z} C_1 + P_1 e^{-\eta_1 z} C_2 + P_2 e^{\eta_2 z} C_3 - P_2 e^{-\eta_2 z} C_4, \\ \sigma_{zz} = -P_3 e^{\eta_1 z} C_1 - P_3 e^{-\eta_1 z} C_2 - P_4 e^{\eta_2 z} C_3 - P_4 e^{-\eta_2 z} C_4, \end{cases} \tag{16}$$

where $P_1 = i\mu\left(\gamma^2 + \eta_1^2\right)$, $P_2 = 2\mu\eta_2\gamma$, $P_3 = 2\mu\eta_1\gamma$, $P_4 = i\left((\lambda + 2\mu)\eta_2^2 - \lambda\gamma^2\right)$. The following boundary conditions hold for layered media:

1. Continuity of displacement and stress:

$$u_x(z_j)\big|_{z=z_j+0} = u_x(z_j)\big|_{z=z_j-0}, \tag{17}$$

$$u_z(z_j)\big|_{z=z_j+0} = u_z(z_j)\big|_{z=z_j-0}, \tag{18}$$

$$\sigma_{xz}(z_j)\big|_{z=z_j-0} = \sigma_{xz}(z_j)\big|_{z=z_j+0}, \tag{19}$$

$$\sigma_{zz}(z_j)\big|_{z=z_j-0} = \sigma_{zz}(z_j)\big|_{z=z_j+0}. \tag{20}$$

2. On the Earth's surface, the stress tensor vanishes and expressions (5), (6) take the form

$$\begin{cases} \sigma_{xz}\big|_{z=0} = 0, \\ \sigma_{zz}\big|_{z=0} = 0. \end{cases} \tag{21}$$

3. As $z \to \infty$, we have

$$u_x(z) = 0, \quad u_z(z) = 0. \tag{22}$$

We have to find

$$u_x(z=0) = u_x^0, \quad u_z(z=0) = u_z^0.$$

In each layer, the solution of a system of two second-order equations with constant coefficients depends on four unknown constants, and in the half-space $z > z_N$ it depends on two constants, i.e., in total, we have to determine $4N-2$ unknown constants. We have four continuity conditions on each boundary z_j, $j \in [1,N]$, and two boundary conditions on the Earth's surface $z_0 = 0$, i.e., a total of $4N-2$ equations. Assuming that the determinant of the system vanishes, we determine the constants from these equations and thus find the amplitude of the traveling wave at any point in the layered medium. Equating the determinant to zero, we obtain an equation that produces the traveling-wave propagation constant γ for the given medium and at the given field frequency ω. This approach is extremely complicated. The calculations can be simplified by using the impedance tensor of a traveling wave in a layered medium.

Determination of the Impedance Tensor

The impedance tensor of a seismic traveling wave is defined by the linear relationships of the stresses σ_{xz}, σ_{zz} with the displacements u_x, u_z:

$$\begin{cases} \sigma_{xz} = Z_{xx}u_x + Z_{xz}u_z, \\ \sigma_{zz} = Z_{zx}u_x + Z_{zz}u_z, \end{cases} \tag{23}$$

where

$$\hat{Z} = \begin{vmatrix} Z_{xx} & Z_{xz} \\ Z_{zx} & Z_{zz} \end{vmatrix} \tag{24}$$

$\hat{Z}$ is the impedance tensor of rank 2. The impedance tensor is continuous on the discontinuities of λ, μ, and ϱ because the displacements and the stresses are continuous.

Once we have determined the impedance tensor for $z = 0$, i.e., $\hat{Z}(z=0) = \hat{Z}^0$, then from the boundary conditions (19) and (20) at $z = 0$ we have

$$\sigma_{xz}(z=0) = Z^0_{xx}u_x(z=0) + Z^0_{xz}u_z(z=0) = 0,$$
$$\sigma_{zz}(z=0) = Z^0_{zx}u_x(z=0) + Z^0_{zz}u_z(z=0) = 0. \tag{25}$$

For the existence of a traveling wave, i.e., for the existence of non-zero $u_x(0)$ and $u_z(0)$, we should have

$$\det \hat{Z}^0 = Z^0_{xx}Z^0_{zz} - Z^0_{xz}Z^0_{zx} = 0. \tag{26}$$

This is a dispersion equation for the determination of the traveling-wave propagation constant γ as a function of the frequency ω and the parameters of the layered medium.

If the roots of the equation exist, then for a given γ we can find from (1.14) the ratio of the displacement amplitudes in the traveling wave:

$$\frac{u_x(z=0)}{u_x(z=0)} = -\frac{Z^0_{xz}}{Z^0_{xx}} = -\frac{Z^0_{zz}}{Z^0_{zx}}. \tag{27}$$

Thus, everything reduces to the determination of the impedance tensor at $z=0$. Let us derive the equation system for the impedance tensor. To this end, we obtain from (1.12) the z-derivatives of the displacements u_x, u_z:

$$\begin{cases} \dfrac{du_x}{dz} = \dfrac{1}{\mu}Z_{xx}(z)u_x(z) + \left(\dfrac{1}{\mu}Z_{xz}(z) - i\gamma\right)u_z(z), \\ \dfrac{du_z}{dz} = \dfrac{1}{\lambda+2\mu}\left(Z_{zx}(z) - i\gamma\lambda\right)u_x(z) + \dfrac{1}{\lambda+2\mu}Z_{zz}(z)u_z(z). \end{cases} \tag{28}$$

Then, inside the layer, where λ, μ are constant, we find the second derivatives of the displacements:

$$\frac{d^2u_x}{dz^2} = \frac{1}{\mu}\left(Z'_{xx}(z)u_x(z) + Z_{xx}(z)u'_x(z) + Z'_{xz}(z)u_z(z) + Z_{xz}(z)u'_z(z)\right) - i\gamma u'_z(z), \tag{29}$$

$$\frac{d^2u_z}{dz^2} = \frac{1}{\lambda+2\mu}\left(Z'_{zz}(z)u_z(z) + Z_{zz}(z)u'_z(z) + Z'_{zx}(z)u_x(z) + Z_{zx}(z)u'_x(z) - i\gamma\lambda u'_x(z)\right). \tag{30}$$

Substituting in these expressions $u'_x(z)$ and $u'_z(z)$ from (28), we finally obtain

$$\frac{d^2u_x}{dz^2} = \left(\frac{1}{\mu}Z'_{xx} + \frac{1}{\mu^2}Z^2_{xx} + \frac{Z_{xz}Z_{zx} - i\gamma\lambda Z_{xz}}{\mu(\lambda+2\mu)} - \frac{\gamma^2\lambda + i\gamma Z_{zx}}{(\lambda+2\mu)}\right)u_x$$
$$+\left(\frac{Z_{xz}Z_{xx} - i\mu\gamma Z_{xx}}{\mu^2} + \frac{1}{\mu}Z'_{xz} + \frac{Z_{xz}Z_{zz}}{\mu(\lambda+2\mu)} - \frac{i\gamma Z_{zz}}{\lambda+2\mu}\right)u_z, \tag{31}$$

$$\frac{d^2u_z}{dz^2} = \left(\frac{Z'_{zx}}{\lambda+2\mu} + \frac{Z_{zz}Z_{zx} - i\gamma\lambda Z_{zz}}{(\lambda+2\mu)^2} + \frac{Z_{zx}Z_{xx}}{\mu(\lambda+2\mu)} - \frac{i\gamma\lambda Z_{xx}}{\mu(\lambda+2\mu)} \right) u_x$$

$$+ \left(\frac{Z'_{zz}}{\lambda+2\mu} + \frac{Z_{zz}^2}{(\lambda+2\mu)^2} + \frac{Z_{zx}Z_{xz} - i\mu\gamma Z_{zx}}{\mu(\lambda+2\mu)} - \frac{\lambda\mu\gamma^2 + i\gamma\lambda Z_{xz}}{\mu(\lambda+2\mu)} \right) u_z. \tag{32}$$

Substituting the first and second derivatives from (28)–(32) in the Lame equation (2), we obtain

$$\Big(Z'_{xx} - \Big(i\lambda\mu\gamma(Z_{xz} - Z_{zx}) - (\lambda+2\mu)Z_{xx}^2 - \mu Z_{xz}Z_{zx}$$

$$+ 4\mu^2\gamma^2(\lambda+\mu) - \omega^2\rho\mu(\lambda+2\mu)\Big)\Big) u_x + \Big(Z'_{xz} - \Big((\lambda+2\mu) i\mu\gamma Z_{xx}$$

$$- i\mu\lambda\gamma Z_{zz} - (\lambda+2\mu) Z_{xx}Z_{xz} - \mu Z_{xz}Z_{zz}\Big)\Big) u_z = 0, \tag{33}$$

$$\Big(Z'_{zx} + \Big((\lambda+2\mu) i\mu\gamma Z_{xx} - i\gamma\mu\lambda Z_{zz} + (\lambda+2\mu) Z_{zx}Z_{xx} + \mu Z_{zx}Z_{zz}$$

$$- (\lambda+2\mu)\mu^2\gamma^2 + (\lambda+2\mu)\rho\mu\omega^2\Big)\Big) u_x + \Big(Z_{zz} + \Big((\lambda+2\mu) i\mu\gamma Z_{xz}$$

$$- (\lambda+2\mu) i\mu\gamma Z_{zx} + (\lambda+2\mu) Z_{xz}Z_{zx} + \mu Z_{zz}^2 + (\lambda+2\mu)\mu^2\gamma^2\Big)\Big) u_z = 0. \tag{34}$$

Equality to zero should hold for all u_x, u_z, and the parenthetical expressions multiplying u_x and u_z therefore should vanish. As a result, we obtain a system of differential equations for the impedance-tensor components:

$$\begin{cases} \dfrac{dZ_{xx}}{dz} = i\lambda\mu\gamma(Z_{xz} - Z_{zx}) - (\lambda+2\mu)Z_{xx}^2 - \mu Z_{xz}Z_{zx} + 4\mu^2\gamma^2(\lambda+\mu) - \omega^2\rho\mu(\lambda+2\mu), \\ \dfrac{dZ_{xz}}{dz} = -\Big((\lambda+2\mu) i\mu\gamma Z_{xx} - i\mu\lambda\gamma Z_{zz} - (\lambda+2\mu) Z_{xx}Z_{xz} - \mu Z_{xz}Z_{zz}\Big), \\ \dfrac{dZ_{zx}}{dz} = -\Big((\lambda+2\mu) i\mu\gamma Z_{xx} - i\gamma\mu\lambda Z_{zz} + (\lambda+2\mu) Z_{zx}Z_{xx} + \mu Z_{zx}Z_{zz}\Big), \\ \dfrac{dZ_{zz}}{dz} = -\Big((\lambda+2\mu) i\mu\gamma Z_{xz} - (\lambda+2\mu) i\mu\gamma Z_{zx} + (\lambda+2\mu) Z_{xz}Z_{zx} + \mu Z_{zz}^2 + \omega^2\rho\mu(\lambda+2\mu)\Big). \end{cases} \tag{35}$$

The system is solved with given $\{\lambda_n\}$, $\{\mu_n\}$, $\{\rho_n\}$ in each layer; γ is found for each frequency ω. We thus compute $\hat{Z}(z=0) = \hat{Z}^0(\gamma)$ as a function of γ. To solve the system, we need the initial value of the impedance tensor for $z = z_N = H$ (the half-space surface). Given $Z(z=H) = \hat{Z}^H$, we can easily solve numerically the Cauchy problem for $\hat{Z}(z)$.

Experimental Results and Calculation of the Initial Value of $\hat{Z}_H$

Consider the lower half-space for the layered medium $z \in [z_N, \infty)$. There are no reflected waves in this case, and the displacements u_x, u_z are thus

$$\begin{cases} u_x{}^{(N)} = -i\eta_1{}^{(N)} e^{-\eta_1{}^{(N)} z_N} C_1 + \gamma e^{-\eta_2{}^{(N)} z_N} C_2, \\ u_z{}^{(N)} = \gamma e^{-\eta_1{}^{(N)} z_N} C_1 + i\eta_2{}^{(N)} e^{-\eta_2{}^{(N)} z_N} C_2. \end{cases} \tag{36}$$

Differentiating system (36) with respect to z, we obtain

$$\begin{cases} u_x'{}^{(N)} = i\eta_1^{2(N)} e^{-\eta_1{}^{(N)} z_N} C_1 - \eta_2{}^{(N)} \gamma e^{-\eta_2{}^{(N)} z_N} C_2, \\ u_z'{}^{(N)} = -\eta_1{}^{(N)} \gamma e^{-\eta_1{}^{(N)} z_N} C_1 - i\eta_2^{2(N)} e^{-\eta_2{}^{(N)} z_N} C_2. \end{cases} \tag{37}$$

Solving system (36) for C_1 and C_2, we obtain

$$\begin{cases} C_1 = \dfrac{i\eta_2 u_x - u_z \gamma}{\eta_1 \eta_2 - \gamma^2} e^{\eta_1 z}, \\ C_2 = -\dfrac{i\eta_1 u_z + u_x \gamma}{\eta_1 \eta_2 - \gamma^2} e^{\eta_2 z}. \end{cases} \tag{38}$$

Substituting these C_1 and C_2 in (37), jointly with (28) we obtain

$$\begin{cases} (Z_{xx} - W_1) u_x + (Z_{xz} - W_2) u_z = 0, \\ (Z_{zx} - W_3) u_x + (Z_{zz} - W_4) u_z = 0. \end{cases} \tag{39}$$

The parenthetical expressions vanish, and we thus obtain

$$\begin{cases} Z_{xx} = W_1, \\ Z_{xz} = W_2, \\ Z_{zx} = W_3, \\ Z_{zz} = W_4, \end{cases} \tag{40}$$

where

$$W_1 = -\frac{\mu\eta_2\left(\eta_1^2-\gamma^2\right)}{\eta_1\eta_2-\gamma^2}, \quad W_2 = -\frac{i\mu\gamma\left(\eta_1^2-2\eta_1\eta_2+\gamma^2\right)}{\eta_1\eta_2-\gamma^2},$$

$$W_3 = (\lambda+2\mu)\frac{i\gamma\left(\eta_1\eta_2-\eta_2^2\right)}{\eta_1\eta_2-\gamma^2}+i\gamma\lambda, \quad W_4 = (\lambda+2\mu)\frac{\eta_1\left(\eta_2^2-\gamma^2\right)}{\eta_1\eta_2-\gamma^2}.$$

By continuity of the impedance tensor on the boundary $z_N = z_H = H$, we have

$$\hat{Z}_H = \begin{vmatrix} Z_{xx}^{(H)} & Z_{xz}^{(H)} \\ Z_{zx}^{(H)} & Z_{zz}^{(H)} \end{vmatrix}. \tag{41}$$

Each element is a tensor of rank 2 representable in the form

$$Z_{xx}^{(H)} = W_1(\lambda_H,\mu_H,\rho_H,\gamma), \quad Z_{xz}^{(H)} = W_2(\lambda_H,\mu_H,\rho_H,\gamma),$$

$$Z_{zx}^{(H)} = W_3(\lambda_H,\mu_H,\rho_H,\gamma), \quad Z_{zz}^{(H)} = W_4(\lambda_H,\mu_H,\rho_H,\gamma).$$

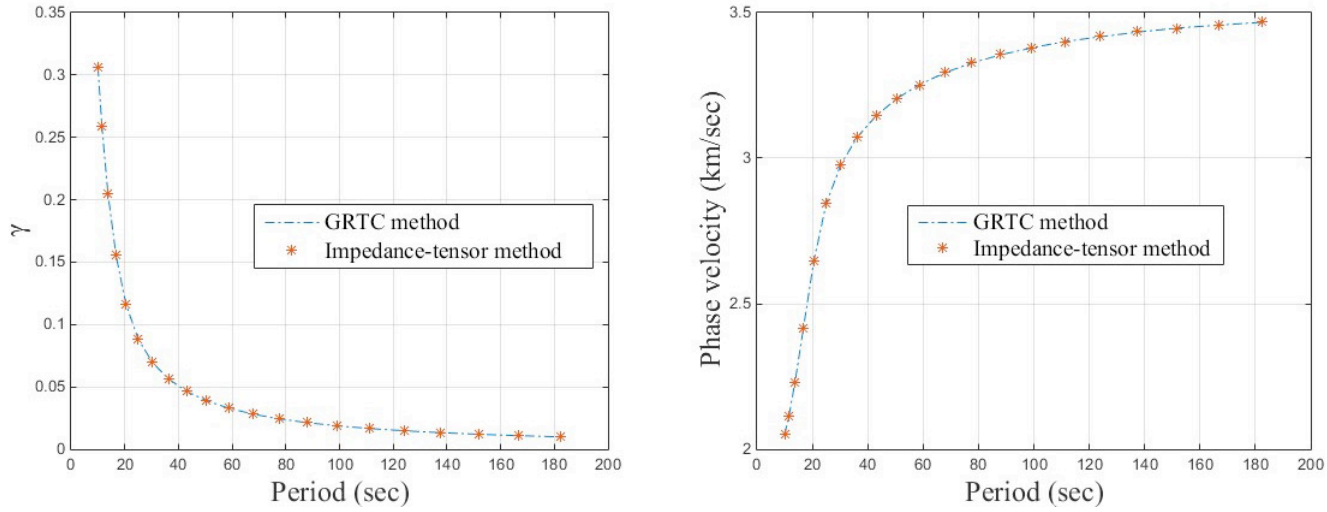
We have thus obtained the initial values of the system of differential equations (35) in terms of the constants λ_H, μ_H, ρ_H, :

$$\gamma\begin{cases} Z_{xx}^{(H)} = -\dfrac{\mu^{(H)}\eta_2{}^{(H)}\left(\eta_1^{2(H)}-\gamma^2\right)}{\eta_1{}^{(H)}\eta_2{}^{(H)}-\gamma^2}, \\ Z_{xz}^{(H)} = -\dfrac{i\mu^{(H)}\gamma\left(\eta_1^{2(H)}-2\eta_1{}^{(H)}\eta_2{}^{(H)}+\gamma^2\right)}{\eta_1{}^{(H)}\eta_2{}^{(H)}-\gamma^2}, \\ Z_{zx}^{(H)} = \left(\lambda^{(H)}+2\mu^{(H)}\right)\dfrac{i\gamma\left(\eta_1{}^{(H)}\eta_2{}^{(H)}-\eta_2^{2(H)}\right)}{\eta_1{}^{(H)}\eta_2{}^{(H)}-\gamma^2}+i\gamma\lambda^{(H)}, \\ Z_{zz}^{(H)} = \left(\lambda^{(H)}+2\mu^{(H)}\right)\dfrac{\eta_1{}^{(H)}\left(\eta_2^{2(H)}-\gamma^2\right)}{\eta_1{}^{(H)}\eta_2{}^{(H)}-\gamma^2}. \end{cases} \tag{42}$$

Applying the fourth-order Runge–Kutta method, we obtain the solution $\hat{Z}_0$ of system (35), and from the dispersion equation (26) we find the traveling-wave characteristics for various frequencies $\gamma(\omega)$:

$$f(\gamma) = \det\hat{Z}^0 = Z_{xx}^0 Z_{zz}^0 - Z_{xz}^0 Z_{zx}^0. \tag{43}$$

Since the analytical solution for $\gamma(\omega)$ is difficult to find, we define the corresponding range $\gamma\in[\tilde{\gamma}_1,\tilde{\gamma}_N]$ and evaluate $f(\gamma)$ (the physical values of the characteristic $\gamma(\omega)$ have the form $\gamma(\omega)=\omega/c$, where c is



(a) **(b)**

Fig. 1. (a) Comparison of traveling-wave characteristics γ obtained by the impedance-tensor method and the GRTC method; (b) Comparison of the phase velocities obtained by the impedance-tensor method and the GRTC method.

Table 1. The Coefficients of the Medium in Model 1

Layer	Depth (km)	ρ (g, cm^3)	*Vp* (km/sec)	*Vs* (km/sec)
1	10	2.4	3.5	2.1
2	20	2.5	4	2.7
3	∞	3	6	4

Comments: *Vp* — longitudinal-wave velocity, *Vs* — transverse-wave velocity.

the phase velocity. To obtain the range for γ, we specify the range $c \in [\xi \min(Vs_j), \xi \max(Vs_j)]$ $j = 1, \ldots, N$, $0.8 < \xi < 1$, where Vs_j, Vp_j respectively are the transverse and longitudinal wave velocities in layered media and j is the layer index).

We now specify a simple experimental model for the layered medium (Table 1). Given the constants ρ, *Vp*, *Vs*, we easily compute numerically the Lame coefficients λ, μ for each layer. Substituting all the constant coefficients in (35), (42), and (43), we obtain the characteristics γ and the phase velocities c as a function of the period (Figs. 1). In seismic data processing, we typically use a normal-mode traveling wave of frequency not less than 1 Hz. In numerical experiments, we accordingly take the frequency interval 10–200 sec to calculate the dispersion curve of the normal-mode traveling wave.

To test the stability of the method, we calculated the dispersion curve by two classical methods (Table 2). Model 2 contains a slow layer. Model 3 assumes growth of the wave velocity. Having determined $\gamma(\omega)$, $c(\omega)$, we calculate the group velocities U_γ from the formula

$$U_\gamma = c(\omega)\left(1 + \frac{T}{c(\omega)}\frac{dc(\omega)}{dT}\right)^{-1}, \tag{44}$$

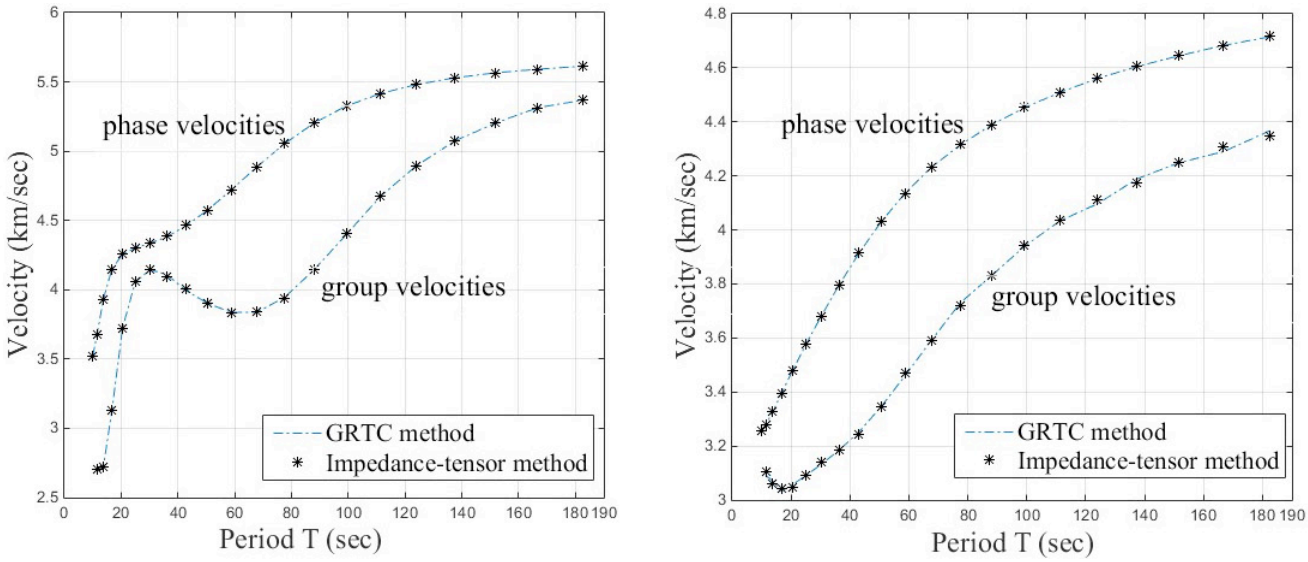


Fig. 2. (a) Comparison of the calculation results obtained with model 2; (b) Comparison of the calculation results obtained with model 3.

Table 2. The Coefficients of the Medium in Models 2 and 3

Layer	Depth (km)		ρ (g, cm^3)		*Vp* (km/sec)		*Vs* (km/sec)	
	M2	M3	M2	M3	M2	M3	M2	M3
1	19	20	2.74	2.8	6.14	6.0	3.55	3.5
2	69	30	3.69	2.9	9.15	6.3	5.04	3.65
3	94	55	3.39	4.0	7.93	6.5	4.37	3.8
4	194	75	4.01	3.2	9.88	7.6	5.45	4.5
5	∞	∞	4.63	2.4	11.35	9.3	6.32	5.6

Comments: M2, M3 stand for model 2 and model 3, respectively.

where T is the traveling-wave period. The results obtained with the impedance-tensor method closely fit those of the GRTC method, with errors of the order of 1% (Fig. 2).

The impedance-tensor method is thus useful for studying the traveling-wave dispersion, as it unambiguously separates the normal-mode traveling wave. This conclusion establishes the efficiency of the proposed method.

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